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GATE ECE · ELECTROMAGNETICS · CHAPTER-WISE PYQS

Waveguides

GATE ECE Electromagnetics · Previous Year Questions with Answer Key

44	32	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	11 questions
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Answer key	at the end

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Marks 1 11 questions

Q1. GATE ECE 2026

The cutoff frequency (in GHz) for the dominant TE_{10} mode of an air-filled rectangular waveguide of inner dimension 0.28 inch $\times 0.14$ inch is ____ .

(rounded off to two decimal places).

Open solution

Q2. GATE ECE 2014 Set 3

Consider an air filled rectangular waveguide with a cross-section of $5\ cm\ \times\ 3\,cm$. For this waveguide, the cut-off frequency (in MHz) of TE_{21} mode is _____

Open solution

Q3. GATE ECE 2011

The modes in a rectangular waveguide are denoted by TE_{mn}/TM_{mn} where m and n are the eigen numbers along the larger and smaller dimensions of the waveguide respectively. Which one of the following satements is TRUE?

(A) The TM_{10} mode of the waveguide does not exist

(B) The TE_{10} mode of the waveguide does not exist

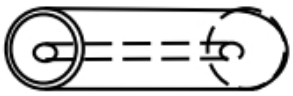
(C) The TM_{10} and the TE_{10} modes both exist and have the same cut-off frequencies

(D) The TM_{10} and the TM_{01} modes both exist and have the same cut-off frequencies.

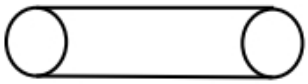
Open solution

Q4. GATE ECE 2009

Which of the following statements is true regarding the fundamental mode of the metallic waveguides shown?

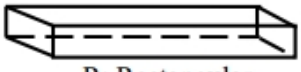


P: Coaxial



Q: cylindrical

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R: Rectangular

(A) Only P has no cutoff- frequency

(B) Only Q has no cutoff-frequency

(C) Only R has no cutoff-frequency

(D) All three have cutoff-frequencies

Open solution

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Q5. GATE ECE 2004

The phase velocity of an electromagnetic wave propagating in a hollow metallic rectangular waveguide in the TE_{10} mode is

(A) equal to its group velocity
(B) less than the velocity of light in free space
(C) equal to the velocity of light in free space
(D) greater than the velocity of light in free space

Open solution

Q6. GATE ECE 2002

The phase velocity for the TE_{10} mode in an air-filled rectangular waveguide is

(A) less than c
(B) equal to c
(C) greater than c
(D) none of the above

Open solution

Q7. GATE ECE 2001

The phase velocity of waves propagating in a hollow metal waveguide is

(A) greater than the velocity of light in free space
(B) less than the velocity of light in free space
(C) equal to the velocity of light in free space
(D) equal to the group velocity

Open solution

Q8. GATE ECE 2001

The dominant mode in a rectangular waveguide is TE_{10}, because this mode has

(A) no attenuation
(B) no cut-off
(C) no magnetic field component
(D) the highest cut-off wavelength

Open solution

Q9. GATE ECE 1999

Indicate which one of the following modes do NOT exist in a rectangular resonant cavity.

(A) TE_{110} (B) TE_{011} (C) TM_{110} (D) TM_{111}

Open solution

Q10. GATE ECE 1997

A rectangular air-filled waveguide has a cross section of $4\,cm \times 10cm$. The minimum frequency which can propagate in the waveguide is

(A) 1.5 GHz (B) 2.0 GHz (C) 2.5 GHz (D) 3.0 GHz

Open solution

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Q11. GATE ECE 1994

The interior of a $\frac{20}{3}\ cm \times \frac{20}{4}\ cm$ rectangular waveguide is completely filled with a dielectric of $\in_r = 4$. Waves of free space wave-lengths shorter than .....can be propagated in the TE_{11} mode.

Open solution

Marks 2 20 questions

Q12. GATE ECE 2022

A waveguide consists of two infinite parallel plates (perfect conductors) at a separation of 10^{-4} cm, with air as the dielectric. Assume the speed of light in air to be 3×10^8 m/s. The frequency/frequencies of TM waves which can propagate in this waveguide is/are _____.

(A) 6×10^{15} Hz (B) 0.5×10^{12} Hz (C) 8×10^{14} Hz (D) 1×10^{13} Hz

Open solution

Q13. GATE ECE 2021

A standard air-filled rectangular waveguide with dimensions $a = 8\ \text{cm}, b = 4\ \text{cm}$, operates at 3.4 GHz . For the dominant mode of wave propagation, the phase velocity of the signal is v_p. The value (rounded off to two decimal places) of v_p/c, where c denotes the velocity of light, is ____ .

Open solution

Q14. GATE ECE 2017 Set 2

Standard air filled rectangular waveguides of dimensions a = 2.29cm and b = 1.02cm are designed for radar applications. It is desired that these waveguides operate only in the dominant TE_{10} mode with the operating frequency at least 25% above the cut-off frequency of the TE_{10} mode but not higher than 95% of the next higher cutoff frequency. The range of the allowable operating frequency f is

(A) $8.19\ GHz \leq f \leq 13.1\ GHz$ (B) $8.19\ GHz \leq f \leq 12.45\ GHz$
(C) $6.55\ GHz \leq f \leq 13.1\ GHz$ (D) $1.64\ GHz \leq f \leq 10.24\ GHz$

Open solution

Q15. GATE ECE 2016 Set 3

Consider an air-filled rectangular waveguide with dimensions a = 2.286cm and b = 1.016cm. At 10GHz operating frequency, the value of the propagation constant (per meter) of the corresponding propagation mode is _____

Open solution

Q16. GATE ECE 2016 Set 3

Consider an air-filled rectangular waveguide with dimensions a = 2.286 cm and b = 1.016 cm. The increasing order of the cut-off frequencies for different modes is

(A) $TE_{01} < TE_{10} < TE_{11} < TE_{20}$ (B) $TE_{20} < TE_{11} < TE_{10} < TE_{01}$
(C) $TE_{10} < TE_{20} < TE_{01} < TE_{11}$ (D) $TE_{10} < TE_{11} < TE_{20} < TE_{01}$

Open solution

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Q17. GATE ECE 2015 Set 2

An air-filled rectangular waveguide of internal dimension $a\ cm\ \times\ b\ cm$ (a > b) has a cutoff frequency of 6 GHz for the dominant TE_{10} mode. For the same waveguide, if the cutoff frequency of the TE_{11} mode is 15 GHz, the cutoff frequency of the TE_{01} mode in GHz is _____

Open solution

Q18. GATE ECE 2015 Set 1

The longitudinal component of the magnetic field inside an air-filled rectangular waveguide made of a perfect electric conductor is given by the following expression
$H_z(x,\ y,\ z,\ t) = 0.1\ \cos(25\ \pi x)\ \cos(30.3\pi y)\ \cos(12\pi\ \times\ 10^9\ t\ -\beta z)(A/m)$
The cross-sectional dimemsions of the waveguide are given as a = 0.08 m and b = 0.033 m. The mode of propagation inside the waveguide is

Open solution

(A) TM_{12} (B) TM_{21} (C) TE_{21} (D) TE_{12}

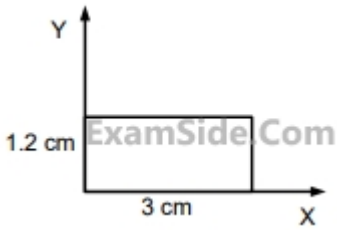
Q19. GATE ECE 2014 Set 2

For a rectangular waveguide of internal dimensions $a\ \times\ b$ (a > b), the cut-off frequency for the TE_{11} mode is the arithmetic mean of the cut-off frequencies for TE_{10} mode and TE_{20} mode. If a $= \sqrt{5}\,cm$, the value of b (in cm) is

Open solution

Q20. GATE ECE 2012

The magnetic field along the propagation direction inside a rectangular waveguide with the cross section shown in the figure is $H_Z = 3\ \cos\ (2.094\ \times\ 10^2 x)\ \cos(2.618\ \times\ 10^2 y)$
$\cos\ (6.283\ \times\ 10^{10} t\ -\beta z)$
The phase velocity V_p of the wave inside the waveguide satisfies



Open solution

(A) V_p > c (B) V_p = c (C) 0 < V_p < c (D) V_p = 0

Q21. GATE ECE 2008

A rectangular waveguide of internal dimensions (a = 4 cm and b = 3 cm) is to be operated in TE_{11} mode. The minimum operating frequency is

Open solution

(A) 6.25 GHz (B) 6.0 GHz (C) 5.0 GHz (D) 3.75 GHz

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Q22. GATE ECE 2007

An air-filled rectangular waveguide has inner dimensions of $3\,cm \times 2\,cm$. The wave impedance of the TE_{20} mode of propagation in the waveguide at a frequency of 30 GHz is (free space impedance $\eta_0 = 377\ \Omega$)

Open solution

(A) 308 Ω (B) 355 Ω (C) 400 Ω (D) 461 Ω

Q23. GATE ECE 2007

The $\vec{E}$ field in a rectangular waveguide of inner dimensions $a \times b$ is given by
$\vec{E} = \frac{\omega\mu}{h^2}\left(\frac{\pi}{a}\right) H_0 \sin\left(\frac{2\pi x}{a}\right) \sin(\omega t - \beta z)\hat{y}$,
where H_0 is a constant, a and b are the dimensions along the x-axis and the y-axis respectively. The mode of propagation in the waveguide is

Open solution

(A) TE_{20} (B) TM_{11} (C) TM_{20} (D) TE_{10}

Q24. GATE ECE 2006

A rectangular waveguide having TE_{10} mode as dominant mode is having a cutoff frequency of 18 GHz for the TE_{30} mode. The inner broad-wall dimension of the rectangular waveguide is

Open solution

(A) 5/3 cms (B) 5 cms (C) 5/2 cms (D) 10 cms

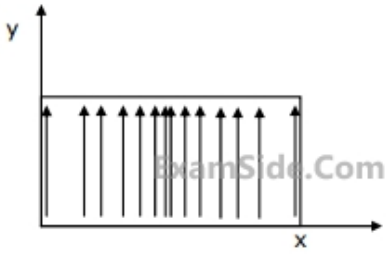
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Q25. GATE ECE 2005

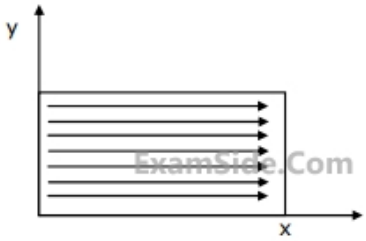
Which one of the following does represent the electric field lines for the TE_{02} mode in the cross-section of a hollow rectangular metallic waveguide?

Open solution

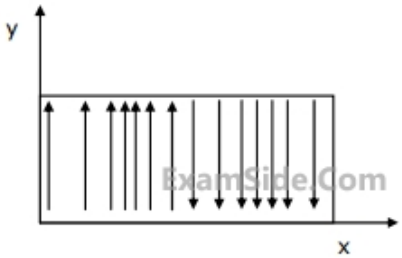
(A)



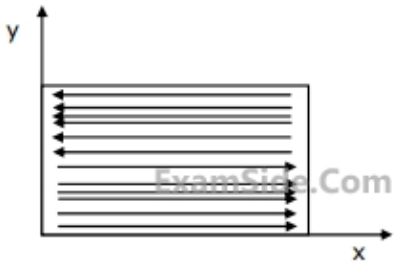
(B)



(C)



(D)



Q26. GATE ECE 2003

A rectangular metal wave guide filled with a dielectric material of relative permittivity $\varepsilon_r = 4$ has the inside dimensions $3.0\,cm \times 1.2\,cm$. The cut-off frequency for the dominant mode is

(A) 2.5 GHz (B) 5.0 GHz (C) 10.0 GHz (D) 12.5 GHz

Open solution

Q27. GATE ECE 2000

A rectangular waveguide has dimensions $1\ cm \times 0.5\ cm$. Its cut-off frequency is

(A) 5 GHz (B) 10 GHz (C) 15 GHz (D) 20 GHz

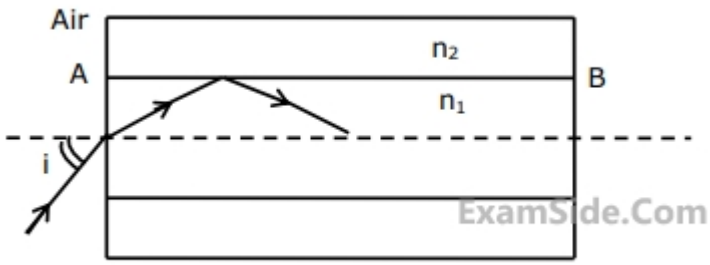
Open solution

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Q28. GATE ECE 1993

Open solution

An optical fiber consists of a cylindrical dielectric rod of refractive index n_1, surrounded by another dielectric of refractive index n_2 where $n_2 < n_1$ as shown in the following Fig. If a ray is incident from air at angle i to the axis, then it undergoes total internal reflection at the interface AB if



(A) $i \geq \sin^{-1}\sqrt{{n_1}^2 - {n_2}^2}$

(B) $i < \sin^{-1}\sqrt{n_1 - n_2}$

(C) $i \leq \sin^{-1}\sqrt{{n_1}^2 - {n_2}^2}$

(D) $i = \sin^{-1}\sqrt{n_1 - n_2}$

Q29. GATE ECE 1990

Open solution

Choose the correct statements.
For a wave propagating in an air filled rectangular wave guide

(A) Guided wavelength is never less than the free space wavelength

(B) Wave impedance is never less than the free space impedance.

(C) Phase velocity is never less than the free space velocity

(D) TEM mode is possible if the dimensions of the wave guide are properly chosen.

Q30. GATE ECE 1988

Open solution

For a normal mode EM wave propagating in a hollow rectangular wave guide

(A) the phase velocity is greater than the group velocity

(B) the phase velocity is greater than velocity of light in free space

(C) the phase velocity is less than the velocity of light in free space

(D) the phase velocity may be either greater than or less than the group velocity

Q31. GATE ECE 1987

Open solution

The cut-off frequency of a waveguide depends upon

(A) The dimensions of the waveguide

(B) The dielectric property of the medium in the waveguide

(C) The characteristic impedance of the waveguide

(D) The transverse and axial components of the fields

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Marks 5 10 questions

Q32. GATE ECE 2002

Open solution

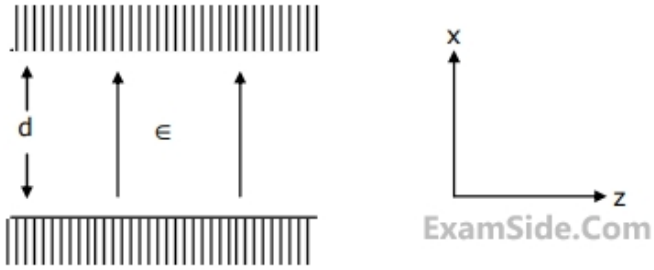
Consider a parallel plate waveguide with plate separation d as shown in Fig. The electric and magnetic fields for the TEM mode are given by $E_x = E_0\, e^{-j\,k\,z+j\,\omega\,t}$, $H_y = \frac{E_0}{\eta}\, e^{-j\,k\,z+j\,\omega\,t}$

Where $k = \eta\omega \in$

(a) Determine the surface charge densities ρ_s on the plates at x = 0 and x = d

(b) Determine the surface current densities $\overrightarrow{J_s}$ on the same plates.

(c) Prove that ρ_s and $\overrightarrow{J_s}$ satisfy the current continuity condition.



Q33. GATE ECE 2002

Open solution

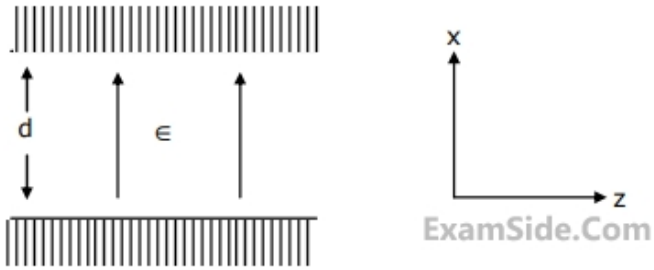
Consider a parallel plate waveguide with plate separation d as shown in Fig. The electric and magnetic fields for the TEM mode are given by $E_x = E_0\, e^{-j\,k\,z+j\,\omega\,t}$, $H_y = \frac{E_0}{\eta}\, e^{-j\,k\,z+j\,\omega\,t}$

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Q34. GATE ECE 2002

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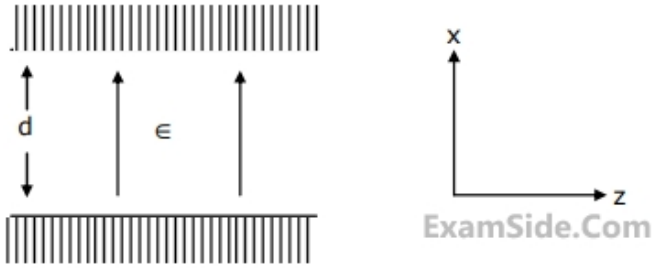
Consider a parallel plate waveguide with plate separation d as shown in Fig. The electric and magnetic fields for the TEM mode are given by $E_x = E_0\, e^{-j\,k\,z+j\,\omega\,t}$, $H_y = \frac{E_0}{\eta}\, e^{-j\,k\,z+j\,\omega\,t}$

Where $k = \eta\omega \in$

(a) Determine the surface charge densities ρ_s on the plates at x = 0 and x = d

(b) Determine the surface current densities $\overrightarrow{J_s}$ on the same plates.

(c) Prove that ρ_s and $\overrightarrow{J_s}$ satisfy the current continuity condition.



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Q35. GATE ECE 2001

Open solution

A rectangular hollow metal waveguide has dimensions a = 2.29 cm and b = 1.02 cm. Microwave power at 10 GHz is transmitted through the waveguide in the TE_{10} mode.

(a) Calculate the cut-off wavelength and the guide wavelength for this mode.

(b) What are the other (TE or TM) modes that can propagate through the waveguide?

(c) If a = b = 2.29 cm, what are the modes which can propagate through the waveguide?

Q36. GATE ECE 1999

Open solution

A 100 m section of an air-filled rectangular wave-guide operating in the TE_{10} mode has a cross-sectional dimension of 1.071 cm $\times$ 0.5 cm. Two pulses carriers of 21 GHz and 28 GHz are simultaneously launched at one end of the wave-guide section. What is the time delay difference between the two pulses at the other end of the waveguide?

Q37. GATE ECE 1998

Open solution

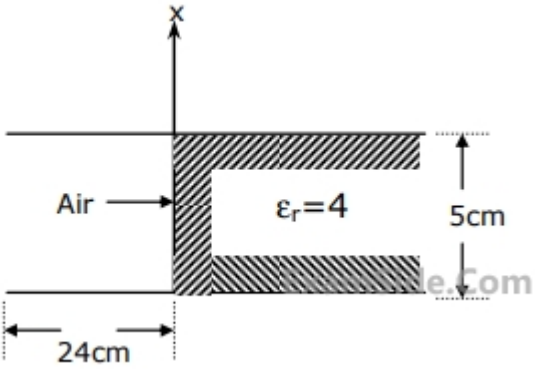
A rectangular waveguide with inner dimensions 6 cm $\times$ 3 cm has been designed for a single mode operation. Find the possible frequency range of operation such that the lowest frequency is 5% above the cut off and the highest frequency is 5% below the cut off of the next higher mode.

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Q38. GATE ECE 1998

The region between a pair of parallel perfectly conducting planes of infinite extent in the y and z directions is partially filled with a dielectric as shown in Figure. A 30 GHz TE_{10} wave is incident on the air dielectric interface as shown. Find the VSWR at the interface.

Open solution



Q39. GATE ECE 1996

In an air-filled rectangular waveguide, the vector electric field is given by
$\overrightarrow{E} = \cos(20\pi y)\ \exp\ \left[-j\left(\frac{40\pi}{3}\right)z + j\omega t\right]\hat{i}\ V/m$
Find the vector magnetic field and the phase velocity of the wave inside the waveguide.

Open solution

Q40. GATE ECE 1995

A rectangular hollow metal waveguide is required to be so designed to propagate a 9375 MHz signal in its TE_{10}-mode that the guide-wavelength equals the cut-off wavelength. Calculate the value of 'a' (breadth or the wider dimension of the waveguide). Take b = a/2. Also, calculate the cut-off frequency of the next higher order mode.

Open solution

Q41. GATE ECE 1994

A rectangular hollow metal waveguide of internal cross-section of $7.366\ cm\ \times\ 3.556\,cm$ carries a 3 GHz signal in the TE_{10}-mode. Calculate the maximum power handling capability of the waveguide assuming the maximum permissible electric field inside the waveguide to be 30 kV/cm

Open solution

Marks 8 2 questions

Q42. GATE ECE 1988

A rectangular waveguide with cross-section dimensions 5 cm by 3 cm is filled with a dielectric of relative permittivity 3.
(i) Determine the cutoff frequency of theTE_{11} mode.
(ii) Determine the frequency at which this / mode has an attenuation of 3π nepers/m.

Open solution

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Q43. GATE ECE 1988

A rectangular waveguide with cross-section dimensions 5 cm by 3 cm is filled with a dielectric of relative permittivity 3.
(i) Determine the cutoff frequency of theTE_{11} mode.
(ii) Determine the frequency at which this / mode has an attenuation of 3π nepers/m.

Open solution

Marks 10 1 questions

Q44. GATE ECE 1989

Design a rectangilar hollow metal waveguide to operate in the X-band such that at an operating free-space wavelength of 3.2 cm
(a) The guide-wavelength for the TE_{10^-} mode is 40% more than the operating free-space wavelength.
(b) The cut-off wavelength for the TM_{11^-} mode is 40% of the cut-off wavelength for the TE_{10^-} mode.

Open solution

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Answer Key

Marks 1

1. **(21.09) (20.50 to 21.50)**	2. **11.7**	3. **A**	4. **A**	5. **D**
6. **C**	7. **A**	8. **D**	9. **A**	10. **A**
11. **8 cm**				

Marks 2

12. **A, C**	13. **1.15 to 1.20**	14. **B**	15. **158**	16. **C**
17. **13.75**	18. **C**	19. **2**	20. **D**	21. **A**
22. **C**	23. **A**	24. **C**	25. **D**	26. **A**
27. **C**	28. **C**	29. **A, C**	30. **A, B**	31. **A, B**

Marks 5

32. **(a)** $P_s = \varepsilon\ E_0 \cos(wt - kz)$. **(b)** $\overrightarrow{J_s} = -\frac{E_0}{\eta} \cos(wt - kz)$	33. **(a)** $P_s = \varepsilon\ E_0 \cos(wt - kz)$. **(b)** $\overrightarrow{a_s}\overrightarrow{J_s} = -\frac{E_0}{\eta} \cos(wt - kz)$	34. **(a)** $P_s = \varepsilon\ E_0 \cos(wt - kz)$. **(b)** $\overrightarrow{a_s}\overrightarrow{J_s} = -\frac{E_0}{\eta} \cos(wt - kz)$	35. **(a) cut-off wavelength = 4.58 cm Guide wavelength = 4 cm (b) only TE_{10} mode can propagate. (c) TE_{10}, TE_{11}, TE_{01}, TM_{11} can propagate.**	36. **Time delay differance = 0.04 μS**
37. **Range = 2.625 GHz - 4.75 GHz**	38. **VSWR = 2**	39. **solve it**	40. **a = 2.26 cutoff frequency = 13257 MHz**	41. **23 MW**

Marks 8

42. **(i) 3.36 GHz (ii) 3.35 GHz**	43. **(i) 3.36 GHz (ii) 3.35 GHz**

Marks 10

44. **solve it**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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